



MANCHESTER TECHNOLOGIES COLLEGE

GRAND TEST – NEET

TOTAL MARKS – 720 MARKS

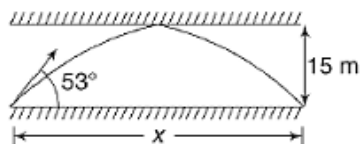
DURATION – 3.00 HOURS

DATE – 16-07-2026

PART A – PHYSICS

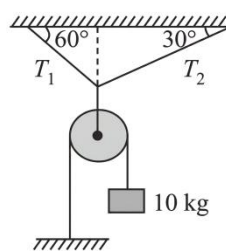
- A body is thrown vertically upwards with a velocity u . Select the incorrect statements from the following.
 - Both velocity and acceleration are zero at its highest point
 - Velocity is maximum and acceleration is zero at the highest point.
 - Velocity is maximum and acceleration is g downwards at its highest point.
- A ball is shot from the floor in a long hall having a roof at a height of 15 m. Initial velocity of the ball is 25 ms^{-1} at an angle of 53° with the floor. The ball lands on the floor at a distance x m from the point of projection. Assume collision with roof as elastic, if any, and find

- 20 m
- 40 m
- 30 m
- 15 m



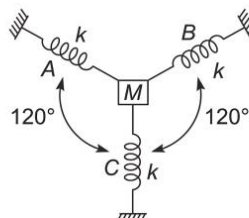
- In the arrangement as shown, tension T_2 is ($g = 10 \text{ m/s}^2$)

- 50 N
- 100 N
- $50\sqrt{3}$ N
- $100\sqrt{3}$ N

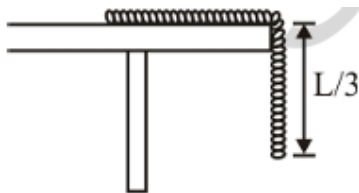


- The speed of travelling wave whose quantities is given by $y = 2 / [(x - 3t)^2 + 1]$, (all variables are in SI unit) is
 - 3 m/s
 - $1/3$ m/s
 - 9 m/s
 - $1/9$ m/s
- Two simple pendulums of lengths 9m and 25m, are both given small displacements in the same direction at the same instant. If they are in phase after the shorter pendulum has completed n oscillations, n is equal to :-
 - 5
 - $5/2$
 - $3/2$
 - 3
- A block of mass m is connected to three springs, each of spring constant k as shown in the block is pulled by x in the direction of C. Find resultant spring constant.

- k
- $2k$



- (3) $3k$
 (4) $3K/2$
7. A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1 . When it is projected with velocity u at an angle $(\pi/2 - \theta)$ with the horizontal, it reaches maximum height H_2 . The relation between the horizontal range R of the projectile, H_1 and H_2 is.
 (1) $R = 4\sqrt{H_1 H_2}$ (2) $R = 4(H_1 - H_2)$ (3) $R = 4(H_1 + H_2)$ (4) $R = H_1^2 / H_2^2$
8. The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is
 (1) $2 : 3$ (2) $2 : 1$ (3) $\sqrt{5} : \sqrt{6}$ (4) $1 : \sqrt{2}$
9. The moment of inertia of two rotating bodies A and B are I_A and I_B ($I_A > I_B$). If their angular momentum are equal, then
 (1) Kinetic energy of A = Kinetic energy of B
 (2) Kinetic energy of $A >$ Kinetic energy of B
 (3) Kinetic energy of $A <$ Kinetic energy of B
 (4) Kinetic energy of the two bodies cannot be compared with the given data
10. A diatomic gas ($\gamma = 1.4$) does 200 J of work when it is expanded isobarically. Find the heat given to the gas in the process-
 (1) 400 J (2) 500 J (3) 700 J (4) 300 J
11. A gas is compressed isothermally to half of its initial volume, The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then-
 (1) Compressing the gas through adiabatic process will require more work to be done.
 (2) Compressing the gas isothermally or adiabatically will require the same amount of work.
 (3) Which of the case (Whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.
 (4) Compressing the gas isothermally will require more work to be done.
12. There are 100 divisions on the circular scale of a screw gauge of pitch 1 mm. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found the 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is :
 (1) 4.65 mm (2) 4.55 mm (3) 4.60 mm (4) 3.35 mm
13. A silver wire has mass (0.6 ± 0.006) g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be
 (1) 4% (2) 3% (3) 6% (4) 7%
14. A circular disc of radius R is removed from a bigger circular disc of radius $2R$ such that the circumference of the disc coincide. The center of mass of the new disc is αR from the center of the bigger disc. The value of α is :-
 (1) $1/3$ (2) $1/2$ (3) $1/6$ (4) $1/4$
15. A stone is tied to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u . The magnitude of the change in its velocity as it reaches a position where the string is horizontal is :-
 (1) $\sqrt{u^2 - 2gL}$ (2) $\sqrt{2gL}$ (3) $\sqrt{u^2 - gL}$ (4) $\sqrt{2(u^2 - gL)}$
16. A uniform chain (mass M , length L) is released from rest from a smooth horizontal surface as shown in the figure. Velocity of the chain at the instant it completely comes out of the table will be:-



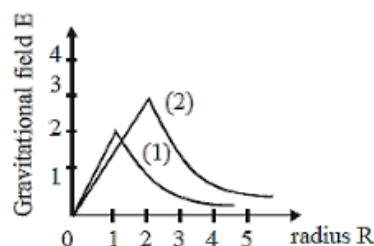
- (1) $v = \sqrt{gL}$ (2) $v = \sqrt{3gL/4}$ (3) $v = \sqrt{8gL/9}$ (4) $v = \sqrt{7gL/8}$

17. The escape velocity from a planet is V_e . A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

- (1) v_e (2) $v_e/\sqrt{2}$ (3) $v_e/2$ (4) Zero

18. Consider two solid spheres of radii $R_1 = 1\text{m}$, $R_2 = 2\text{m}$ and masses M_1 and M_2 , respectively. The gravitational field due to sphere (1) and (2) are shown. The value of M_1/M_2 is:

- (1) $1/2$
(2) $2/3$
(3) $1/3$
(4) $1/6$



19. Two wires, one of steel and the other of brass having same lengths and radii in the ratio 1:2 are suspended from the same roof. What should be the ratio of weights suspended from steel and brass wires so that the lower ends of both the wires are at the same level? (Given $Y_{\text{steel}} = 2Y_{\text{brass}}$)

- (1) 2 : 1 (2) 4 : 1 (3) 1 : 2 (4) 1 : 4

20. A large number of liquid drops each of radius a coalesce to form a single spherical drop of radius b . The energy released in the process is converted into kinetic energy of the big drop formed. The speed of big drop will be :- (T = surface tension, d = density of liquid)

- (1) $\sqrt{[(6T/d)(1/a - 1/b)]}$ (2) $\sqrt{[(4T/d)(1/a - 1/b)]}$
(3) $\sqrt{[(8T/d)(1/a - 1/b)]}$ (4) $\sqrt{[(5T/d)(1/a - 1/b)]}$

21. A stone tied to the end of a 80 cm long string is whirled in a horizontal circle with a constant speed. If the stone makes 14 revolutions in 25 s, the magnitude of its acceleration is.

- (1) 20 m/s^2 (2) 12 m/s^2 (3) 9.9 m/s^2 (4) 8 m/s^2

22. A point initially at rest moves along x-axis. Its acceleration varies with time as $a = (6t + 5) \text{ m/s}^2$. If it starts from origin, the distance covered in 2 s is

- (1) 20 m (2) 18 m (3) 16 m (4) 25 m

23. A block of mass m is placed on a surface with a vertical cross section given by $y = x^3/6$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is:

- (1) $1/6 \text{ m}$ (2) $2/3 \text{ m}$ (3) $1/3 \text{ m}$ (4) $1/2 \text{ m}$

24. In a series resonant circuit, having L, C and R as its elements, the resonant current is i . The power dissipated in circuit at resonance is:

- (1) $i^2 R / (\omega L - 1/\omega C)$ (2) Zero (3) $i^2 \omega L$ (4) $i^2 R$

25. A circular wire of radius r rotates about its own axis with angular speed ω in a magnetic field B perpendicular to its plane, then the induced e.m.f. is :

- (1) $\frac{1}{2} B r \omega^2$ (2) $B r^2$ (3) $2 B r \omega^2$ (4) Zero

26. Select the wrong statement. EM waves :

- (1) Are transverse in nature.

- (2) Travel in free space at a speed of light.
- (3) Are produced by accelerating charges
- (4) Travel in all media with same speed.

27. An electric charge $+q$ moves with velocity $\vec{v} = 3\hat{i} + 4\hat{j} + \hat{k}$ in an electromagnetic field given by $\vec{E} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 3\hat{k}$. The y-component of the force experienced by $+q$ is :

- (1) $11q$
- (2) $5q$
- (3) $3q$
- (4) $2q$

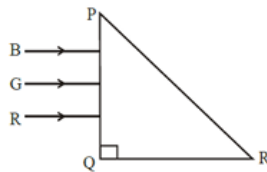
28. A charged particle moves with velocity $\vec{v} = a\hat{i} + d\hat{j}$ in a magnetic field $\vec{B} = A\hat{i} + D\hat{j}$. The force acting on the particle has magnitude F . Then :

- (1) $F=0$, if $a D=d A$
- (2) $F=0$, if $a D=-d A$
- (3) $F=0$, if $A a=-d D$
- (4) $F \propto (a^2 + b^2)^{1/2} \times (A^2 + D^2)^{1/2}$

29. A conducting circular loop of radius r carries a constant current i . It is placed in a uniform magnetic field B such that B is perpendicular to the plane of the loop. The magnetic force acting on the loop is:

- (1) irB
- (2) $2\pi r i B$
- (3) Zero
- (4) $\pi r i B$

30. Three rays of light, namely red (R), green (G) and blue (B) are incident on the face PQ of a right angled isosceles prism PQR as shown in figure.



The refractive indices of the material of the prism for red, green and blue wavelengths are 1.27, 1.42 and 1.49 respectively. The colour of the ray(s) emerging out of the face PR is :

- (1) Green
- (2) Red
- (3) Blue and green
- (4) Blue

31. An object is placed a distance of 40 cm from a concave mirror of focal length 15 cm . If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be-

- (1) 30 cm away from the mirror
- (2) 36 cm away from the mirror
- (3) 30 cm towards the mirror
- (4) 36 cm towards the mirror

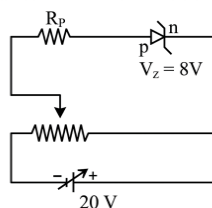


The logic circuit shown above is equivalent to:

- (1)
- (2)
- (3)
- (4)

33. A Zener diode having Zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across Zener diode is as shown in the diagram. The value of protective resistance R_p is ____ Ω

- (1) 19.2
- (2) 192
- (3) 1920
- (4) 1.92



34. An X-ray tube is operated at 1.24 million volt. The shortest wavelength of the produced photon will be :

- (1) 10^{-3} nm
- (2) 10^{-1} nm
- (3) 10^{-2} nm
- (4) 10^{-4} nm

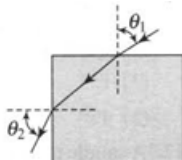
35. Consider the following statements:

- A. Atoms of each element emit characteristics spectrum.
- B. According to Bohr's Postulate, an electron in a hydrogen atom, revolves in a certain stationary orbit.
- C. The density of nuclear matter depends on the size of the nucleus.
- D. A free neutron is stable but a free proton decay is possible.
- E. Radioactivity is an indication of the instability of nuclei.

Choose the correct answer from the options given below :

- (1) A, B, C, D and E
- (2) A, B and E only
- (3) B and D only
- (4) A, C and E only

36. Light is incident on a glass block as shown in figure.



If θ_1 is increased slightly then what happens to θ_2

- (1) θ_2 also increases slightly
- (2) θ_2 unchanged
- (3) θ_2 decreases slightly
- (4) θ_2 changes abruptly since the ray experiences total internal reflection.

37. Two polaroids are crossed to each other. When one of them is rotated through 60° then what percentage of the incident unpolarised light will be transmitted by the polaroids ?

- (1) 50%
- (2) 75%
- (3) 62.5%
- (4) 37.5%

38. The objective lens of a compound microscope produces magnification of 10. In order to get an overall magnification of 100 when image is formed at 25 cm from the eye, the focal length of the eye lens should be

- (1) 4 cm
- (2) 10 cm
- (3) 25/9 cm
- (4) 9 cm

39. Which one of the following statement is false?

- (1) Pure Si doped with trivalent impurities gives a p-type semiconductor.
- (2) Majority carriers in a n-type semiconductor are holes.
- (3) Minority carriers in a p-type semiconductor are electrons.
- (4) The resistance of intrinsic semiconductor decreases with increase of temperature.

40. An electric dipole is placed along x axis in an electric field $E = x^2\hat{i}$, the dipole will

- (1) Undergo rotational motion only
- (2) Translate only along x axis
- (3) Rotate and translate along z axis
- (4) Rotate and translate along x axis

41. A charge Q is distributed over two concentric hollow spheres of radii R and r ($R > r$) such that their surface densities are equal. The common potential at the centre is (in cgs system)

- (1) ZERO
- (2) $2kQ[(r + R)/(r^2 + R^2)]$
- (3) $kQ[(r + R)/(r^2 + R^2)]$
- (4) $\frac{1}{2}kQ[(r + R)/(r^2 + R^2)]$

42. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

- A. The charge stored in it, increases.
- B. The energy stored in it, decreases.
- C. Its capacitance increases.
- D. The ratio of charge to its potential remains the same.
- E. The product of charge and voltage increases.

Choose the most appropriate answer from the options given below:

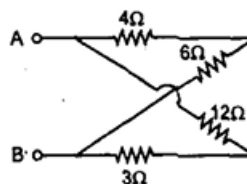
- (1) B, D and E only (2) A, B and C only (3) A, B and E only (4) A, C and E only

43. A voltmeter consists of a 25Ω coil connected in series with a 575Ω resistor. The coil takes 10 mA for full scale deflection. What maximum potential difference can be measured on this voltmeter?

- (1) 3V (2) 6V (3) 0.6V (4) 0.3V

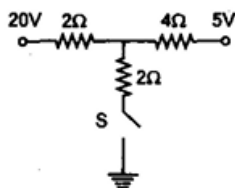
44. In the given network, the equivalent resistance between A and B is

- (1) 6Ω
(2) 16Ω
(3) 7Ω
(4) 5Ω



45. As the switch S is closed in the circuit shown in figure, current passed through it is:

- (1) 4.5 A
(2) 6.0 A
(3) 3.0 A
(4) Zero



PART B – CHEMISTRY

46. Which of the following statement is incorrect.

- (i) Octet expansion of 2nd period elements is not possible.
- (ii) Octet rule dictates that atoms are most stable when their valence shells are filled with 8 electrons.
- (iii) BeCl₂ dimer does not follow octet rule.
- (iv) Formal charge can be calculated with the help of octet rule, for 2nd period elements.
- (v) Hydrogen can never obey octet rule.
- (vi) SF₄, PCl₅ and IF₇ are example of octet expansion.
- (vii) Duplet rule is followed by hydrogen and helium.
- (viii) F₂ is an example of incomplete octet.

- (1) Statement number (ii) (2) Statement number (iv)
(3) Statement number (vi) (4) Statement number (viii)

47. **Statement I** : The bond order of helium molecule (He₂) is zero.

Statement II : The number of electrons in bonding molecular orbital and antibonding molecular orbital are equal.

- (1) If both statement I and statement II are true and the statement II is the correct explanation of the statement I.
- (2) If both statement I and statement II are true but statement II is not the correct explanation of the statement I.
- (3) If statement I is true but statement II is false.
- (4) If the statement I and statement II both are false.

48. A 5.6 g sample of an organic compound containing only C, H and O gave 8.8 g CO₂ and 3.6 g H₂O on complete combustion. If the molar mass of the compound is 56 g/mol, the number of oxygen atoms per molecule is:

- (1) 1 (2) 2 (3) 3 (4) 0

49. When 1.0 L of 0.2 M K₂Cr₂O₇ reacts completely with excess FeSO₄ in acidic medium, the mass of Fe³⁺ formed (in g) is: (Fe = 56)

- (1) 67.2 (2) 33.6 (3) 44.8 (4) 22.4

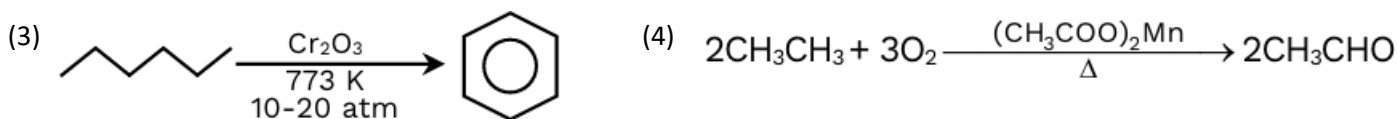
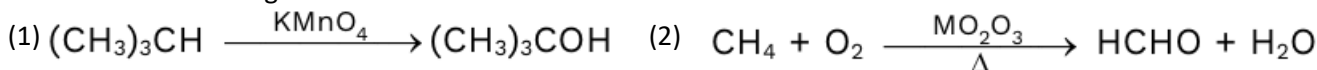
50. Which of the following statement is **wrong**.

- (1) Aldehydes react with one equivalent of monohydric alcohol in presence of dry hydrogen chlorides to yield alkoxyalcohol intermediate, known as acetals.
- (2) NaHSO₃ are useful for separation and purification of aldehyde.

(3) Cyanohydrins are useful synthetic intermediates.

(4) Lucas test can be used for distinguishes of primary, secondary and tertiary alcohol.

51. Which of the following is **incorrect**?



52. The magnetic moment of a certain complex (A) of Co was found to be 4.89 B.M. and the EAN as 36. Co also forms complex (B) with magnetic moment 3.87 B.M. and EAN 37, and complex (C) with EAN as 36 but diamagnetic.

Which of the following statements is true regarding the above observation?

- (1) The oxidation states of Co in (A), (B), and (C) are +3, +2, and +3, respectively
- (2) Complexes (A) and (B) have sp^3d^2 hybridization state while (C) has dsp^3 hybridization state
- (3) The spin multiplicities of Co in (A), (B), and (C) are 4, 3, and 1, respectively
- (4) The oxidation states of Co in (A), (B), and (C) are +6, +8, and +1, respectively

53. Pick the incorrect statement

- (1) Actinoids have higher tendency of complex formation as compared to lanthanoids
- (2) All actinoids are radioactive whereas all lanthanides except promethium are nonradioactive
- (3) Oxides and hydroxides of actinoids are more basic than that of lanthanoids
- (4) Highest oxidation state exhibited by actinoids is lower than that of lanthanoids

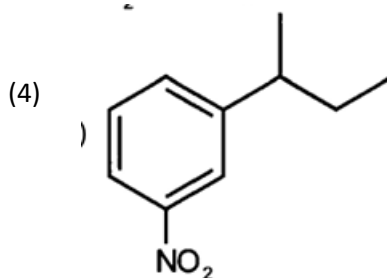
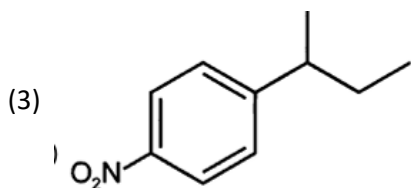
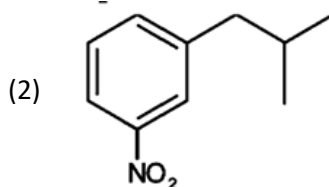
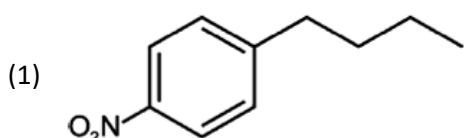
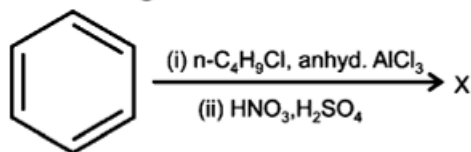
54. The wavelength of the radiation emitted when an electron in a hydrogen atom transitions from the orbit whose angular momentum is $3h/2\pi$ to the orbit with angular momentum h/π is: ($R_H = 1.097 \times 10^7 \text{ m}^{-1}$)

- (1) 6563 Å (2) 4861 Å (3) 1025 Å (4) 1216 Å

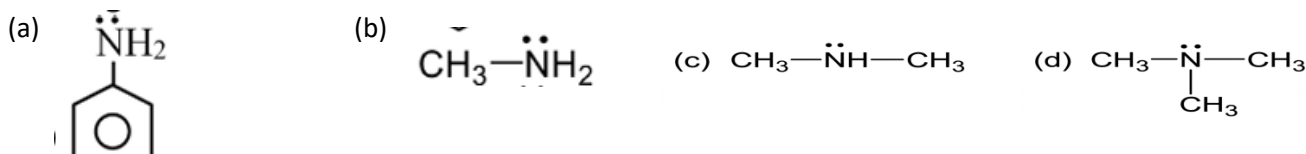
55. Which of the following sets of quantum numbers is NOT permissible?

- (1) $n=3, l=2, m_l=-1, m_s=+\frac{1}{2}$ (2) $n=4, l=3, m_l=+3, m_s=-\frac{1}{2}$
- (3) $n=2, l=2, m_l=0, m_s=+\frac{1}{2}$ (4) $n=5, l=0, m_l=0, m_s=-\frac{1}{2}$

56. The major product 'X' formed in the following reaction is



57. Arrange in correct basic strength order



- (1) $d > c > b > a$ (2) $a > b > c > d$ (3) $c > b > d > a$ (4) $c > d > b > a$

58. Which is not correct statement about the chemistry of 3d and 4f series elements

- (1) 3d-elements show more oxidation states than 4f-series elements
 (2) The energy difference between 3d and 4s orbitals is very little
 (3) Europium(II) is more stable than Ce(II)
 (4) The paramagnetic character in 3d-series elements increases from scandium to copper

59. Identify the correct statement?

- (1) $B < C < N < O$: First ionization enthalpy (2) $S^{2-} > Cl^- > K^+ > Ca^{2+}$: Ionization enthalpy
 (3) $Al^{3+} < Mg^{2+} < Na^+ < F^-$: Ionic size (4) $F > Cl > Br > I$: E.A

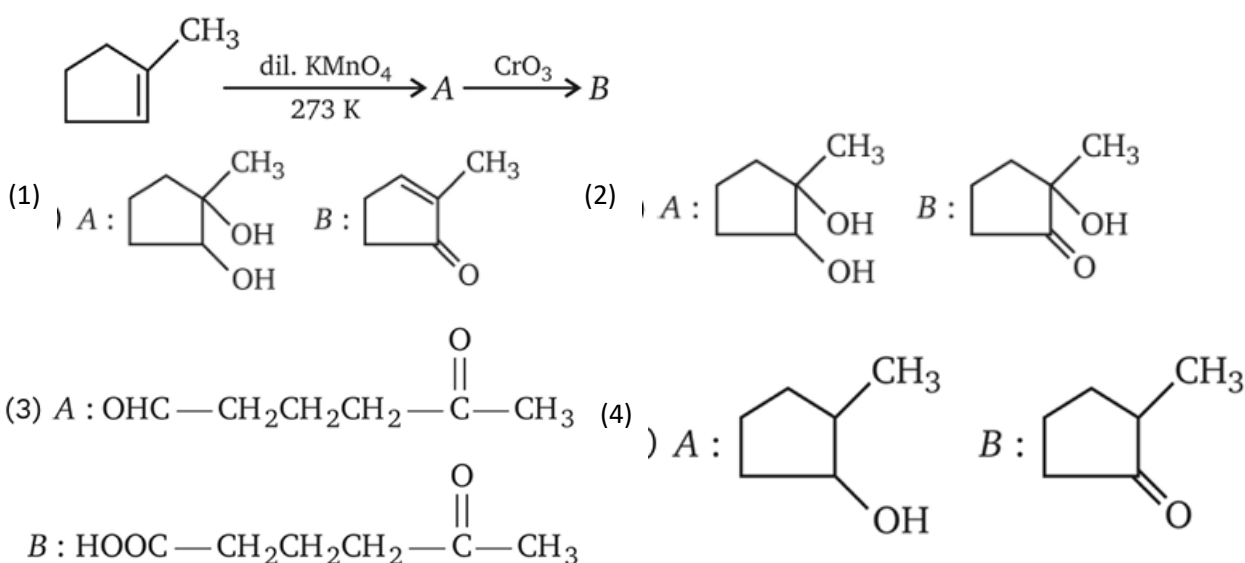
60. For the reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K_c = 4 \times 10^{-4}$ at temperature T. If 1 mol N_2 and 1 mol O_2 are placed in a 1 L vessel, the equilibrium concentration of NO (in M) is approximately:

- (1) 0.02 (2) 0.04 (3) 0.01 (4) 0.2

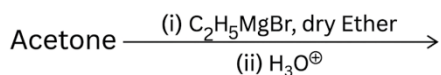
61. The pH of a 0.01 M solution of a weak acid HA ($K_a = 4 \times 10^{-5}$) is closest to:

- (1) 2.7 (2) 3.2 (3) 4.4 (4) 2.0

62. Identify products A and B :



63. What is the IUPAC name of the organic compound formed in the following chemical reaction?



- (1) 2-Methylpropan-2-ol (2) Pentan-2-ol
 (3) Pentan-3-ol (4) 2-Methylbutan-2-ol

64. Consider the following orders:

- (i) $N_2O_3 > N_2O_4 > N_2O_5$: Acidic character
 (ii) $CH_4 < CCl_4 < CF_4$: Electronegativity of central 'C'-atom
 (iii) $Ca^{2+} < Na^+ < S^{2-} < Se^{2-}$: Ionic radius
 (iv) $Ni > Pd > Pt$: Ionisation energy
 (v) $As^{5+} > Sb^{5+} > Bi^{5+}$: Stability of oxidation state
 (vi) $LiF > NaF > KF > RbF$: Lattice energy
 (vii) $Cl > Br > F > I$: Electron affinity

Number of correct orders are :-

(1) 3

(2) 4

(3) 5

(4) 6

65. The correct order of bond enthalpy (kJ mol^{-1}) is ____.

(1) $\text{Si-Si} > \text{C-C} > \text{Sn-Sn} > \text{Ge-Ge}$

(2) $\text{Si-Si} > \text{C-C} > \text{Ge-Ge} > \text{Sn-Sn}$

(3) $\text{C-C} > \text{Si-Si} > \text{Sn-Sn} > \text{Ge-Ge}$

(4) $\text{C-C} > \text{Si-Si} > \text{Ge-Ge} > \text{Sn-Sn}$

66. Equal volumes of 0.1 M CH_3COOH and 0.05 M NaOH are mixed. If K_a for $\text{CH}_3\text{COOH} = 1.8 \times 10^{-5}$, the resulting solution is:

(1) An acidic buffer with $\text{pH} < 4.74$

(2) An acidic buffer with $\text{pH} = 4.74$

(3) A basic buffer with $\text{pH} > 4.74$

(4) A neutral solution

67. **Assertion (A):** The common ion effect reduces the solubility of a sparingly soluble salt.

Reason (R): Addition of a common ion shifts the dissolution equilibrium towards the undissolved salt as per Le Chatelier's principle.

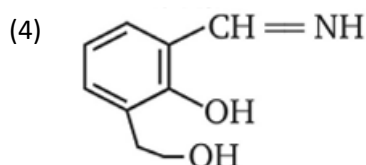
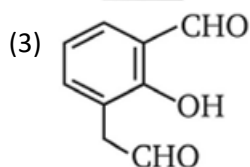
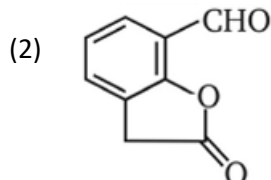
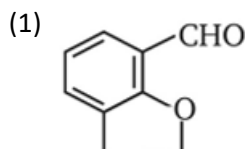
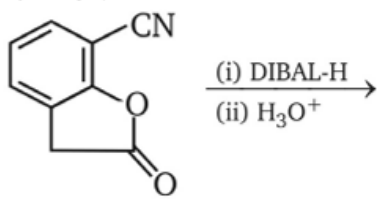
(1) Both A and R are true; R is the correct explanation of A

(2) Both A and R are true; R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

68. The major product of the following reaction is :



69. Match List I with List II

List I – Natural amino acid	List II – One letter code
a Aspartic acid	i Y
b Arginine	ii R
c Tyrosine	iii F
d Phenylalanine	iv D

(1) a-iv, b-ii, c-i, d-iii

(2) a-i, b-ii, c-iv, d-iii

(3) a-ii, b-i, c-iii, d-iv

(4) a-i, b-ii, c-iii, d-iv

70. The correct statement about oxy acids of chlorine

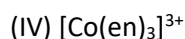
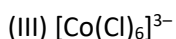
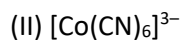
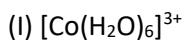
(1) All the oxy acids have same number of un protonated oxygens

(2) In all the oxy acids oxidation state of chlorine is same

(3) All the oxy acids have different number of Cl-OH bonds

(4) Have different number of Cl=O bonds

71. Identify the correct order of wavelength of light absorbed for the following complex ions.



- (1) $\text{III} > \text{I} > \text{IV} > \text{II}$ (2) $\text{II} > \text{IV} > \text{I} > \text{III}$ (3) $\text{III} > \text{I} > \text{II} > \text{IV}$ (4) $\text{I} > \text{III} > \text{IV} > \text{II}$

72. For a reaction at 300 K: $\Delta H = -60 \text{ kJ/mol}$ and $\Delta S = -180 \text{ J/(mol}\cdot\text{K)}$. At what temperature (in K) does the reaction become non-spontaneous?

- (1) Above 333 K (2) Above 500 K (3) Below 333 K (4) At all temperatures

73. Heat of formation of $\text{Al}_2\text{O}_{3(s)}$ is $(-x) \text{ Kcal/mol}$. Calculate heat of combustion of $\text{Al}_{(s)}$ is ?

- (1) $(-x) \text{ Kcal/mol}$ (2) $(-2x) \text{ Kcal/mol}$ (3) $(-x/2) \text{ Kcal/mol}$ (4) Can not be calculated

74. **Assertion** : $(\text{CH}_3)_3\text{C}^+$ is more stable than $\text{CH}_3\text{C}^+\text{H}_2$ and C^+H_3 is the least stable cation.

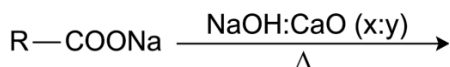
Reason : Hyperconjugation interaction in $(\text{CH}_3)_3\text{C}^+$ is greater than in $\text{CH}_3\text{C}^+\text{H}_2$ as the $(\text{CH}_3)_3\text{C}^+$ has nine C-H bonds. In C^+H_3 , vacant p orbital is perpendicular to the plane in which C-H bonds lie; hence cannot overlap with it. Thus, C^+H_3 lacks hyperconjugative stability

The correct answer will be :-

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion
(2) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion
(3) Assertion is true and Reason is false
(4) Assertion is false and Reason is true

75. Which of the following statements is/are **correct**?

(A) Wurtz reaction is used for the preparation of higher alkanes containing even number of carbon atoms.



(B) $\text{R}-\text{H} + \text{Na}_2\text{CO}_3$. Where x : y is 3 : 1

(C) Ozonolysis reaction is highly useful in detecting the position of the double bond in alkenes or other unsaturated compound

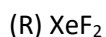
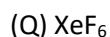
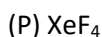
(D) Alkenes on reaction with cold, dilute, aqueous solution of potassium permanganate (**Baeyer's reagent**) produce vicinal glycols.

- (1) Only a, d (2) Only a, b, c (3) Only a, c, d (4) a, b, c, d

76. Among the following metal carbonyl species, the one with the highest metal-carbon back bonding is

- (1) $[\text{Ti}(\text{CO})_6]^{2-}$ (2) $[\text{V}(\text{CO})_6]^-$ (3) $\text{Cr}(\text{CO})_6$ (4) $[\text{Mn}(\text{CO})_6]^+$

77. The compounds which on complete hydrolysis forms XeO_3



(1) P, R

(2) P, Q

(3) Q, R

(4) P, Q, R

78. The osmotic pressure of a 0.1 M solution of a substance that dimerises to 50% extent in solution is: ($T = 300 \text{ K}$, $R = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

- (1) 1.845 atm (2) 2.46 atm (3) 1.23 atm (4) 3.69 atm

79. **Assertion (A)**: A solution of 6 g urea in 100 g water shows the same freezing point depression as a solution of 9 g glucose in 100 g water.

Reason (R): The colligative property depends on the number of moles of solute, not on its nature. (Urea = 60, Glucose = 180)

- (1) Both A and R are true; R is the correct explanation of A
(2) Both A and R are true; R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

80. Using the provided information in the following paper chromatogram :-

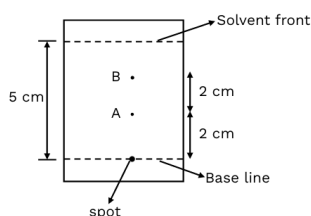


Fig : paper chromatography for compound A and B calculate R_f value of B

- (1) 0.4 (2) 2.5 (3) 1.25 (4) 0.8

81. Match the List-I with List-II.

List-I (Compound)	List-II B.P.(K)
A Propanal	i 370
B Acetone	ii 322
C Propan-1-ol	iii 329
D Methoxyethane	iv 281
E n-Butane	v 273

- (1) A-ii, B-iii, C-i, D-iv, E-v (2) A-iii, B-ii, C-i, D-iv, E-v
 (3) A-i, B-ii, C-iii, D-iv, E-v (4) A-ii, B-i, C-iii, D-v, E-iv

82. The three lanthanoids which exhibit +2 oxidation state are

- (1) Sm, Tb, Gd (2) Sm, Eu, Yb (3) La, Gd, Lu (4) Yb, Pm, Sm

83. Identify the number of correct statements:-

- (a) NaNO_3 & KNO_3 are known as Indian salt-petre and Chile-salt petre respectively
 (b) There is a considerable increase in covalent radius from N to P.
 (c) Bismuth hardly forms any compound in -3 oxidation state
 (d) Except nitrogen, all the elements show allotropy in group 15.
 (e) Single N-N bond is weaker than the single P-P bond
 (f) Magnesium can combine with bismuth to form magnesium bismuthide
 (g) In oxy acids of phosphorus, P-H bonds are not ionisable to give H^+ and do not play any role in basicity
 (h) For group 16, Bonding in +4 and +6 oxidation states are primarily covalent
 (i) Atomic radius of aluminium is 143 pm.
 (j) Due to property of catenation and $p\pi-p\pi$ bond formation, carbon is able to show allotropic forms.

- (1) 6 (2) 7 (3) 8 (4) 9

84. A first-order reaction is 75% complete in 40 minutes. The time required for it to be 99.9% complete (in minutes) is closest to: ($\log 2 = 0.301$)

- (1) 120 (2) 200 (3) 100 (4) 160

85. **Assertion (A):** For a zero-order reaction, the half-life is directly proportional to the initial concentration.

Reason (R): For a zero-order reaction, $t_{1/2} = [\text{A}]_0/2k$.

- (1) Both A and R are true; R is the correct explanation of A
 (2) Both A and R are true; R is NOT the correct explanation of A
 (3) A is true but R is false
 (4) A is false but R is true

86. The number of aromatic compounds from following are:-

- (1) a-ii; b-i; c-iv; d-iii (2) a-iii; b-iv; c-i; d-ii
(3) a-ii; b-iv; c-i; d-iii (4) a-i; b-iv; c-iii; d-ii

PART C – BIOLOGY

91. Read the following term carefully :-

Algin, Carrageen, Mannitol, Starch, Polysiphonia, Dictyota, Fucus, Pear shaped Zoospore, Oogamous, Volvox, Chlorophyll 'd', Fucoxanthin, ribbon shaped chloroplast, holdfast

How many terms are related with brown algae

- (1) 10 (2) 12 (3) 9 (4) 8

92. Read the following statements and find out how many are correct.

- (1) In roots the parenchymatous cells which lie between xylem and phloem are called conjunctive tissue.
(2) All the tissues on the inner side of endodermis such as pericycle, vascular bundles and pith constitute the stele.
(3) When the xylem and phloem are situated at the same radius of vascular bundles it is called conjoint vascular bundle.
(4) In leaves pericycle & endodermis is present

- (1) Three (2) One (3) Four (4) Two

93. Mendel's work remain unrecognised for long time due to :-

- I. Communication was not easy.
II. Concept of factors which did not blend was not accepted
III. Use of mathematics to explain biological problem was unacceptable
IV. He could not provide any physical proof for the existence of factors.

Choose the right combinations

- (1) I and II (2) II and III (3) III and IV (4) All of these

94. Find the incorrect statement:

- (1) In bundle sheath cells of C_4 plants, first product of CO_2 fixation is $3C$ compound.
(2) In photorespiration both chloroplast and peroxisome utilize O_2 .
(3) PEP carboxylase plays important role in synthesis of sugar in mesophyll cells of C_4 plants.
(4) ATP requirement of C_4 plants is higher as compared to C_3 plants to produce one molecule of sugar.

95. A plant with genotype TtRrEE produce 400 gametes. Find out number of gametes which contain tRE genotype?

- (1) 300 (2) 200 (3) 100 (4) 0

96. Match the following:-

Column-I	Column-II
(a) Sickle cell anaemia	(i) Deletion
(b) Thalassemia	(ii) Translocation
(c) CML	(iii) Lethal genetic disorder
(d) Cry du chat	(iv) Substitution

- (1) a-iv, b-iii, c-ii, d-i (2) a-iii, b-iv, c-ii, d-i
(3) a-ii, b-iii, c-i, d-iv (4) a-iii, b-iv, c-i, d-ii

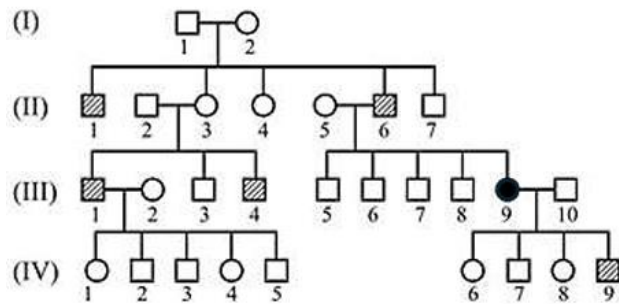
97. Fill in the blanks :-

- (i) The human genome project was completed in ..a.. and the sequence of chromosome 1 was completed in ..b....
(ii) Chromosome 1 hasc.... and chromosome Y has ...d... genes.

- (1) a-1990, b-2006, c-231, d-2968 (2) a-2006, b-2003, c-231, d-2968
(3) a-1990, b-2006, c-2968, d-231 (4) a-2003, b-2006, c-2968, d-231

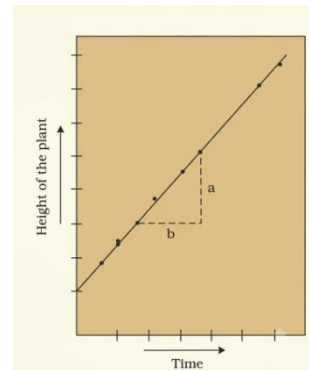
98. Following pedigree represents the inheritance of a rare disorder in an extended family. What is the most likely mode of inheritance for this disease?

- (1) Autosomal recessive
- (2) X-linked recessive
- (3) X-linked dominant
- (4) Y-linked



99. Which of the following statements are correct regarding the type of growth represented by the given diagram ?

- (a) It is observed at early stage of zygotic division
 - (b) Number of dividing cells remain same
 - (c) It represents root growth at tip
 - (d) Represented by expression $W_1 = W_0 e^{rt}$
- (1) a and c
 - (2) b, c and d
 - (3) a, b and d
 - (4) b and c



100. Which one of the following statement is not correct?

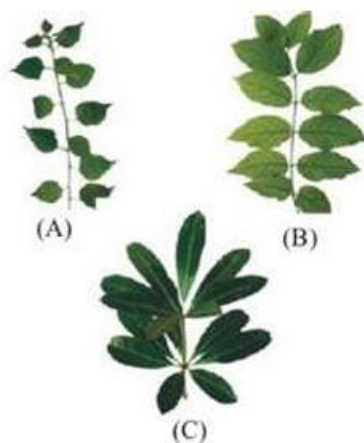
- (1) In a monohybrid cross of Mendel F_1 always resembled either one of the parents.
- (2) Self pollination of tall F_1 plant produces only tall plant.
- (3) Punnett square is a graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross.
- (4) The contrasting traits did not show any blending at either F_1 or F_2 stage in monohybrid cross.

101. Match the Column-I, Column-II and Column-III

Column I	Column II	Column III
A. Cruciferae	I. $Br \oplus \overset{\circ}{P}_{(3+3)} \overset{\circ}{A}_{3+3} \overset{\circ}{G}_{(3)}$	1.
B. Fabaceae	II. $Br \oplus \overset{\circ}{K}_{(5)} \overset{\circ}{C}_{(5)} \overset{\circ}{A}_5 \overset{\circ}{G}_2$	2.
C. Solanaceae	III. $\% \overset{\circ}{K}_{(5)} \overset{\circ}{C}_{1+2+(2)} \overset{\circ}{A}_{(9)+1} \overset{\circ}{G}_1$	3.
D. Liliaceae	IV. $\oplus \overset{\circ}{K}_{2+2} \overset{\circ}{C}_4 \overset{\circ}{A}_{2+4} \overset{\circ}{G}_{(2)}$	4.

- (1) A-IV-2; B-III-1; C-II-3; D-I-4
- (2) 2-A-IV; B-III-2; C-II-3; D-I-4
- (3) 3-A-IV-2; B-III-1; C-II-4; D-I-3
- (4) A-IV-2; B-III-3; C-II-1; D-I-4

102. The given figure shown some types of phyllotaxy select the option that correctly match them.



	A	B	C
(1)	Opposite eg. Calotropis, Guava	Alternate eg. Mustard, Chinarose	Whorled eg. Alstonia
(2)	Alternate eg. Calotropis, Guava	Opposite eg. Chinarose, Mango	Whorled eg. Mustard
(3)	Alternate eg. Mustard, Chinarose	Opposite eg. Calotropis, Guava	Whorled eg. Alstonia
(4)	Opposite eg. Chinarose, Guava	Alternate eg. Calotropis, Mustard	Whorled eg. Sunflower

103. How many statements are correct:

- (a) VNTR belong to class of a minisatellite.
- (b) Satellite DNA do not code for proteins.
- (c) DNA from every cell or tissue from an individual show different degree of polymorphism.
- (d) 99.9 percent DNA of all humans are same

- (1) All statements
- (2) Only one statement
- (3) 2 statements
- (4) 3 statements

104. What is it that forms the basis of DNA Fingerprinting?

- (1) The relative amount of DNA in the ridges and grooves of the fingerprints.
- (2) Satellite DNA occurring as highly repeated short DNA segments
- (3) The relative proportions of purines and pyrimidines in DNA
- (4) The relative difference in the DNA occurrence in blood, skin and saliva

105. a. Number of common characteristic increases from kingdom to species in taxonomic hierachy

b. Order being a higher category exhibits many similar characters than family.

Choose the correct option.

- (1) Both a & b are correct
- (2) Only b is correct
- (3) Only a is correct
- (4) Both a & b are incorrect

106. Read the following statements w.r.t double fertilisation.

- a. PEN develops into endosperm
- b. One male gamete fuse with haploid secondary nucleus
- c. It is an event unique to gymnosperms and angiosperms
- d. One of the male gamete fuses with the diploid secondary nucleus to produce PEN

Mark the option representing incorrect statements

- (1) a, d
- (2) a, b
- (3) c, d
- (4) b, c

107. Angiospermic family including colchicine producing plant

- (1) Bears non-endospermous seeds
- (2) Has tricarpellary syncarpous gynoecium with axile placentation
- (3) Lacks cymose inflorescence
- (4) Is characterised by presence of zygomorphic flowers

108. Meristems which appear early in life of higher plants and are responsible for the formation of the primary plant body are

- a. Interfascicular cambium
- b. Intercalary meristem
- c. Lateral meristem
- d. Phellogen
- e. Apical meristem

- (1) a and b
- (2) b and e
- (3) c and d
- (4) a, c and d

109. a. Bulliform cells

- b. Presence of cuticle
- c. Piliferous layer
- d. Bean-shaped guard cells

Which of the above feature(s) belong(s) to dicot leaves?

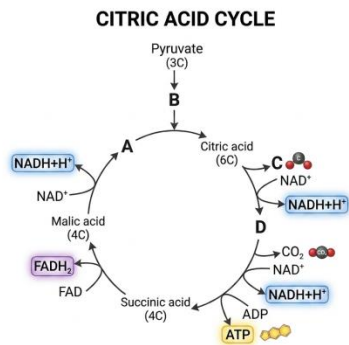
- (1) Only d
- (2) a and c
- (3) b and c
- (4) b and d

110. Choose the incorrect match w.r.t. elements of phloem and their features.

	Feature
(1) Sieve tube element	Peripheral cytoplasm, large central vacuole but lacks a nucleus
(2) Companion cells	Maintain pressure gradient in sieve tubes
(3) Phloem parenchyma	Storage of resins, latex mucilage and food materials

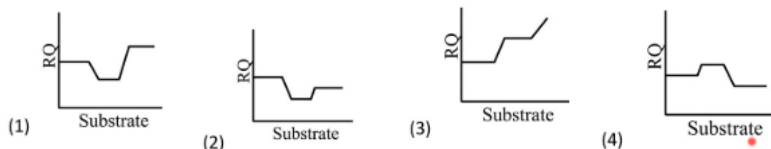
	Feature
(4) Phloem fibers	Generally absent in secondary phloem

111. In the given representation of Krebs cycle, Identify A, B, C and D



	A	B	C	D
(1)	OAA	Acetyl CoA	CO ₂	α-keto-glutaric acid
(2)	OAA	Acetyl CoA	CoA	α-keto glutaric acid
(3)	Acetyl CoA	OAA	CoA	CO ₂
(4)	Acetyl CoA	α-keto-glutaric acid	CO ₂	COA

112. Which of the following graph correctly represents the variation in value of RQ when initially respiratory substrate was carbohydrate, then fats and after that proteins ?



113. A man with blood group B marries a female with blood group A and their first child is having blood group B. What is the genotype of child :-

- (1) I^AI^B
- (2) I^AI^O
- (3) I^BI^O
- (4) I^BI^B

114. Match the Column-I with Column-II.

Column-I	Column-II
(I) Co-dominance	(A) ABO blood group in human being
(II) Multiple Allele	(B) Colour of human being
(III) Incomplete dominance	(C) AB Blood group

Column-I	Column-II
(iv) Polygenic inheritance	(D) Size of starch grain in pea seeds

(1) I- (D); II - (A); III - (C); IV - (B)

(2) I- (D); II - (B); III - (C); IV - (A)

(3) I- (C); II - (A); III - (D); IV - (B)

(4) I- (A); II - (C); III - (B); IV - (D)

115. (I) Fertilization

(II) Liberation of spores

(III) Prothallus formation

(IV) Embryo formation

Arrange the above events in a correct sequence in the life cycle of Pteridophytes :-

(1) II, III, I, IV

(2) IV, III, II, I

(3) I, II, III, IV

(4) I, IV, III, II

116. Match the column-I with column-II and select the correct option from options given below

Column-I	Column-II
(i) Dinoflagellates	(a) Pellicle
(ii) Diatoms	(b) Saprotrophic
(iii) Slime moulds	(c) Cellulosic cell wall
(iv) Euglenoids	(d) Cellulosic cell wall with Silica

(1) i-d, ii-c, iii-b, iv-a

(2) i-c, ii-d, iii-b, iv-a

(3) i-c, ii-d, iii-a, iv-b

(4) i-d, ii-c, iii-a, iv-b

117. A. Sexual reproduction occurs without sex organ

B. Sexual spores are exogenously formed

C. Perfect stage is already discovered

D. Mycelium is branched and septate

Above characters are true for members of

(1) Deuteromycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Phycomycetes

118. Select the correct option to fill up the blanks in the following statements.

a. The(I)..... reproduce by asexual spores called conidia but conidia as well as fruiting bodies are found in(II).....

b. Lichens are very good(III)..... indicators.

- c. The most notorious organism is(IV)..... that causes malaria which has a staggering effect on human population.

	(I)	(II)	(III)	(IV)
(1)	Deuteromycetes	Basidiomycetes	Water pollution	Plasmodium
(2)	Phycomycetes	Ascomycetes	Water pollution	Entamoeba
(3)	Deuteromycetes	Basidiomycetes	Air pollution	Entamoeba
(4)	Deuteromycetes	Ascomycetes	Air pollution	Plasmodium

119. Find out incorrect statement from given below :-

- (1) The main plant body of bryophytes is gamete producing
- (2) In bryophytes spores germinate to form gametophytes
- (3) In gymnosperm the pollen tube release the male gametes in ovule
- (4) After fertilization male gametes develop into embryo and zygote develops in seed

120. Read the following statements :-

- (A) Selaginella and Salvinia are the heterospores pteridophytes because they produce two different kinds of spores.
- (B) The gametophytes of homosporous pteridophytes bear male and female sex organs called antheridia and archegonia respectively
- (C) The female gametophytes in some pteridophytes are retained on the parent sporophytes, for variable period this event is a precursor to seed habit.
- (D) Pteridophytes are first terrestrial plants and they possess seeds

How many of the above statement(s) is/are not correct

- (1) Four
- (2) Three
- (3) Two
- (4) One

121. Complete the below given table by choosing correct option

Type of embryo	Epiblast	Coleorhiza	Root cap
Dicot embryo	A	B	Present
Monocot embryo	Present	Present	C
	A	B	C
(1)	Present	Present	Present
(2)	Absent	Present	Absent

Type of embryo		Epiblast		Coleorhiza		Root cap	
(3)	Absent		Absent		Present		
(4)	Present		Absent		Absent		

122. Which one of the following statements is incorrect?

- (1) Humification leads to accumulation of a dark colored amorphous substance
- (2) Humus is degraded by some microbes and releases the organic nutrients by the process called leaching
- (3) Fragmentation is a process in which detritus is broken down into smaller particles
- (4) Bacterial and fungal enzymes degrade detritus into simpler inorganic substances, by the process called catabolism

123. Select the incorrect statement w.r.t. law of limiting factors.

- (1) It was given by Blackman
- (2) It is applicable when several factors affect any biochemical process
- (3) It helps to identify the factor which directly affects the biochemical process when its quantity is changed
- (4) It states that if a chemical process is affected by more than one factors, then its rate will be determined by the factor which is nearest to its maximum value

124. Match the major causes of biodiversity losses given in column-I with the correct examples given in column-II.

Column-I	Column-II
(a) Habitat loss and fragmentation	(i) Extinction of Steller's sea cow
(b) Over-exploitation	(ii) Carrot grass and Eicchornia
(c) Alien-species invasion	(iii) Affects animal with migratory habits
(d) Co-extinctions	(iv) Coevolved plant-pollinator mutualism

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iii), b(i), c(ii), d(iv)
- (3) a(iv), b(i), c(ii), d(iii)
- (4) a(iii), b(i), c(iv), d(ii)

125. Select the statement which is incorrect w.r.t bacterial transcription

- (1) mRNA does not require any processing to become active
- (2) Transcription and translation take place in separate compartments
- (3) Translation can begin much before the mRNA is fully transcribed
- (4) Transcription and translation can be coupled

126. Match the following columns w.r.t translation and select the correct option

Column-I	Column-II
(a) Charging of tRNA	(i) Binding of amino acid with ATP

Column-I	Column-II
(b) Amino acid activation	(ii) Binding of mRNA with smaller subunit of ribosome
(c) Chain initiation	(iii) Requires ribozymes
(d) Chain elongation	(iv) Transferring of amino acid of tRNA

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(ii), b(iv), c(iii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

127. In Rivet popper hypothesis, the rivets on the wings represents

(1) Key species that drive major ecosystem function

(2) Invasive species that drive major ecosystem function

(3) Ecosystem

(4) Extinct species of an ecosystem

128. Read the following statements and select the correct option.

Statement-I : According to Gause's 'Competitive Exclusion Principle', two closely related species competing for the same resources can co-exist indefinitely.

Statement-II : Majority of parasites reduce the survival, growth and reproduction of the host and reduce its population density.

(1) Only statement-I is correct

(2) Only statement-II is correct

(3) Both the statements are correct

(4) Both the statements are incorrect

129. Mark the incorrect statement w.r.t. predation

(1) Herbivores have a greater advantage since plants cannot run away to avoid predation

(2) If a predator is prudent, the prey population will become extinct

(3) Predators help in maintaining species diversity

(4) Predation keep prey population under control

130. Choose the correct type of ecological pyramid on the basis of following data of dry weights.

(i) Secondary consumer – 11 kg/m²

(ii) Tertiary consumer – 1.5 kg/m²

(iii) Primary producer – 809 kg/m²

(iv) Primary consumer – 37 kg/m²

(1) Upright pyramid of biomass

(2) Inverted pyramid of biomass

(3) Inverted pyramid of energy

(4) Upright pyramid of number

131. Which of the given statement is correct in the context of observing DNA separated by agarose gel electrophoresis?

(1) DNA can be seen in visible light

(2) DNA can be seen without staining in visible light.

- (3) Ethidium bromide stained DNA can be seen in visible light.
 (4) Ethidium bromide stained DNA can be seen under exposure to UV light.

132. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria because of

- (1) insertional inactivation of beta galactosidase in recombinant bacteria.
 (2) Activation of glycosidase enzyme in recombinant bacteria.
 (3) recombinant bacteria containing beta galactosidase.
 (4) Insertional inactivation of alpha galactosidase in non-recombinant bacteria.

133. Read the following statements and select the correct ones :-

- (A) In gel electrophoresis, DNA fragments separate according to their size through sieving effect provided by the agarose gel
 (B) Smallest the fragment size, the farther it moves
 (C) Unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant vector molecule cannot be created.
 (D) Now a days the most commonly used matrix is agarose which is a natural polymer extracted from sea weeds

- (1) A, B, C, D
 (2) A, C
 (3) B, D
 (4) C, D

134. Read the following statement and choose correct one :-

- (1) DNA is hydrophobic molecule, it can pass through cell membrane.
 (2) By micro-injection, recombinant DNA is directly injected into the nucleus of an plant cell
 (3) During DNA isolation we used chilled ethanol in final step
 (4) Retroviruses have been use to deliver desirable gene into plant cells after disarming.

135. Match the following technique and the purpose they are used for

Column-I		Column-II		
(A) DNA probes		(1) To identify cells that have been transformed		
(B) Genetic markers		(2) To join DNA fragments together		
(C) DNA ligase		(3) Complementary DNA, synthesized from mRNA with the help of the enzyme reverse transcriptase		
(D) cDNA		(4) To identify and label a piece of DNA containing a certain sequence using Radioisotope		
	A	B	C	D
(1)	4	1	2	3
(2)	1	4	2	3

Column-I			Column-II	
(3)	1	4	3	2
(4)	4	2	1	3

PART D – ZOOLOGY

136. From the list of given characters, which of them represent Chondrichthyes & Osteichthyes ?

- (a) Placoid scales
- (b) Operculum present
- (c) Air bladder present
- (d) Mostly viviparous
- (e) External fertilisation
- (f) Notochord persistent throughout life.

	Chondrichthyes	Osteichthyes
(1)	a, b, d	c, e, f
(2)	a, d, f	b, c, e
(3)	b, c, d	a, e, f
(4)	b, c, e	a, d, f

137. Metameric segmentation is present in :

- (1) Platyhelminthes, Annelida, Mollusca
- (2) Annelida, Arthropoda, Mollusca
- (3) Annelida, Arthropoda, Chordata
- (4) Enchinodermata, Hemichordata, Chordata

138. Exclusively marine urochordate among the following is

- (1) Salamandra
- (2) Ascidia
- (3) Amphioxus
- (4) Petromyzon

139. There are some features of animals are given :-

Organ system level, Coelomate, Open circulation, Triploblastic, Unsegmented, Intracellular digestion. How many characters are present in the members of phylum arthropoda –

- (1) 3
- (2) 4
- (3) 5
- (4) 2

140. In a frog brain optic lobes are a part of

- (1) Fore brain
- (2) Mid brain
- (3) Hind brain
- (4) Cerebral hemispheres

141. Match column-I with column-II and select correct option :-

Column-I	Column-II

Column-I	Column-II
(a) Sensory papillae	(i) Well organised structure
(b) Nasal epithelium	(ii) Cellular aggregation
(c) Eyes	(iii) Smell
(d) Taste buds	(iv) Organs of touch

(1) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)

(2) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

(3) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(4) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

- 142. Statement-I :** The chloroplast contain chlorophyll and carotenoids which are responsible for trapping light energy essential for photosynthesis.

Statement-II : The matrix of mitochondria possess single, circular DNA molecule, few RNA molecules, ribosome and components required for protein synthesis. In the light of above statements, choose most appropriate answer from the given options.

- (1) Both Statements are correct.
 (2) Both Statements are incorrect.
 (3) Statement-I is correct and Statement-II are incorrect.
 (4) Statement-I is incorrect and Statement-II are correct.

- 143.** Which chromosome has centromere slightly away from the middle of the chromosome resulting into one shorter arm and one longer arm ?

- (1) Metacentric chromosome
 (2) Sub-metacentric chromosome
 (3) Acrocentric chromosome
 (4) Telocentric chromosome

- 144.** Which of the following phenomenon occurs during telophase ?

- (1) Spindle fibres attach to kinetochores of chromosomes.
 (2) Centromeres split and chromatids separate
 (3) Chromatids move to opposite poles.
 (4) Nucleolus, Golgi complex and ER reform.

- 145.** The complete disintegration of __A__ marks the start of __B__ of mitosis, hence the chromosomes are spread through the cytoplasm of the cell. Identify A and B.

- (1) A–Cell membrane, B–Prophase
 (2) A–Nuclear membrane, B–Prophase
 (3) A–Nuclear membrane, B–Metaphase
 (4) A–Cell membrane, B–Metaphase

- 146.** Match the following-

Column-I	Column-II
(a) Prophase-I	(i) Bivalent chromosomes align on equatorial plate

Column-I	Column-II
(b) Metaphase-I	(ii) Homologous chromosome separate
(c) Anaphase-I	(iii) Synaptonemal complex formation
(d) Telophase-I	(iv) Nuclear membrane reappear

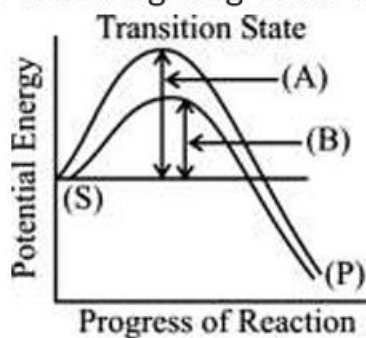
- (1) (a-i), (b-ii), (c-iii), (d-iv)
(2) (a-iii), (b-ii), (c-iv), (d-i)
(3) (a-iii), (b-i), (c-ii), (d-iv)
(4) (a-iv), (b-iii), (c-ii), (d-i)

147. **Statement-I** : All the elements present in a sample of earth crust are also present in a sample of living tissue.

Statement-II : Secondary metabolites have identifiable functions and play known roles in normal physiological processes. Select the correct answer from given below options.

- (1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct and Statement-II is incorrect.
(4) Statement-I is incorrect and Statement-II is correct.

148. In the following diagram A and B is :



- (1) (A) Activation energy without enzyme. (B) Activation energy with enzyme.
(2) (A) Activation energy with enzyme. (B) Activation energy without enzyme.
(3) (A) → Substrate (B) → Product
(4) (A) → Product (B) → Substrate

149. Which is the correct order of cell cycle ?

- (1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow \text{Prophase} \rightarrow \text{Metaphase} \rightarrow \text{Anaphase} \rightarrow \text{Telophase} \rightarrow \text{Cytokinesis}$
(2) $G_2 \rightarrow G_1 \rightarrow S \rightarrow \text{Prophase} \rightarrow \text{Metaphase} \rightarrow \text{Anaphase} \rightarrow \text{Telophase} \rightarrow \text{Cytokinesis}$
(3) $S \rightarrow G_1 \rightarrow G_2 \rightarrow \text{Prophase} \rightarrow \text{Metaphase} \rightarrow \text{Anaphase} \rightarrow \text{Telophase} \rightarrow \text{Cytokinesis}$
(4) $\text{Prophase} \rightarrow \text{Metaphase} \rightarrow \text{Anaphase} \rightarrow \text{Telophase} \rightarrow \text{Cytokinesis} \rightarrow S \rightarrow G_2 \rightarrow G_1$

150. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Anaphase-II begins with the simultaneous splitting of the centromere of each chromosome allowing them to move toward opposite poles of the cell.

Reason (R) : The shortening of microtubules attached to kinetochore takes place during anaphase-II. Choose the correct answer from the options given below :

- (1) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (2) (A) is true but (R) is false.
- (3) (A) is false but (R) is true.
- (4) Both (A) and (R) are true and (R) is correct explanation of (A).

151. Molecule whose molecular weight do not exceed 800 dalton and come under acid insoluble fraction. Above statement is suitable for which biomolecule ?

- (1) Protein
- (2) Polysaccharide
- (3) Lipid
- (4) Nucleic acid

152. Telophase I results in the formation of –

- (1) Two diploid cell
- (2) Two haploid cell
- (3) Four haploid cell
- (4) Four diploid cell

153. In heavy smokers the alveoli of the lungs are enlarged and damaged which reduces the surface area of the exchange of respiratory gases. This condition is called :-

- (1) Asthma
- (2) Silicosis
- (3) Emphysema
- (4) Insomnia

154. Read the following statements :

- (A) Normal heart rate is 72 beats/min.
- (B) Cardiac cycle is complete in 8 sec.
- (C) We can diagnose abnormality of heart by ECG.
- (D) Cholesterol can form stones in kidney.
- (E) RAAS increases BP.
- (F) Aldosterone is secreted by JG cells.

Which of the following is correct statement ?

- (1) A, C and E
- (2) A, B, C and E
- (3) A, B, C, E and F
- (4) B, D and F

155. Which layer of blood vessels is called endothelium ?

- (1) Tunica intima
- (2) Tunica media
- (3) Tunica externa
- (4) All of above

156. Assertion : A minute vessels vasa recta runs parallel to the Henle's loop.

Reason : It play important role in dilution of the urine.

- (1) Both assertion and reason are true but reason is not the correct explanation of assertion.

- (2) Assertion is true but reason is false.
- (3) Assertion is false but reason is true
- (4) Both assertion and reason true and reason is the correct explanation of assertion.

157. Read the following statements

- (a) The oviducts, uterus and vagina constitute the female accessory ducts.
- (b) Each ovary is covered by a thin epithelium which encloses the ovarian stroma.
- (c) The seminiferous tubules of the testis open into the rete testis through vasa efferentia.
- (d) In testis, the immature male germ cells produce sperms.
- (e) Secretion of epididymis, vas deference, seminal vesicle and prostate are essential for maturation and motility of sperms.

Choose the option having all correct statements.

- (1) a, b, c
- (2) a, b, d, e
- (3) b, c, e
- (4) c, d, e

158. Meiosis-I of primary oocyte which takes place during oogenesis in human females is:

- (1) Reductional division with unequal cytokinesis
- (2) Equational division with unequal cytokinesis
- (3) Reductional division with equal cytokinesis
- (4) Equational division with equal cytokinesis

159. After implantation, chorionic villi and uterine tissue become interdigitated with each other to form :

- (1) Chorion and amnion
- (2) Umbilical cord
- (3) Placenta
- (4) Yolk sac

160. Vas deferens receives a duct from __ B __ and opens into urethra as the ejaculatory duct.

Fill in the blank (B) with suitable option.

- (1) B = Seminal vesicle
- (2) B = Bulbourethral gland
- (3) B = Bartholin gland
- (4) B = Prostate gland

161. Which of the following is not a natural contraceptive method?

- (1) Periodic abstinence
- (2) Coitus interruptus
- (3) Lactational amenorrhea
- (4) M.T.P.

162. The medical termination of pregnancy (amendment) act, 2017 was enacted by the government of India with the intension of :

- (1) Reducing the incidence of illegal abortion and consequent maternal mortality and morbidity
- (2) Increasing number of couples with small families
- (3) Better detection and cure of STD
- (4) Increasing number of medically assisted deliveries and better post natal care leading to decreased maternal and infant mortality rates.

163. Which structure is tracts of myelinated nerve fibres?

- (1) Association areas

- (2) Hippocampus
- (3) Corpus callosum
- (4) Amygdala

164. Glucocorticoids stimulates :-

- (A) Gluconeogenesis
- (B) Lipolysis
- (C) Proteolysis

- (1) A only
- (2) A, B and C
- (3) B and C only
- (4) B only

165. Statement-I : Prolonged hyperglycemia leads to a complex disorder diabetes mellitus.

Statement-II : Glucagon is a hypoglycemic hormone while insulin is a hyperglycemic hormone.

- (1) Statement-I and II both are correct.
- (2) Statement-I and II both are incorrect.
- (3) Statement-I is correct but II is incorrect.
- (4) Statement-II is correct but I is incorrect.

166. Match the List-I to List-II :

List-I	List-II
(A) Myosin	(I) Globular protein
(B) G-Actin	(II) Sarcomere
(C) Two adjacent	(III) Meromyosin
(D) Smooth muscles	(IV) Fallopian tube

- (1) A - iii, B - i, C - ii, D - iv
- (2) A - iii, B - ii, C - i, D - iv
- (3) A - ii, B - iii, C - i, D - iv
- (4) A - iv, B - iii, C - ii, D - i

167. Enzymatic activity is destroyed at high temperature because :

- (1) Enzymes are inorganic compounds.
- (2) Proteins are denatured by heat.
- (3) Enzyme becomes more active
- (4) Enzyme permanently binds with substrate.

168. Which of the following is correct name of glycerol ?

- (1) Substituted methane
- (2) Trihydroxy propane
- (3) Triglyceride
- (4) Phosphatidyl choline

169. Read the following statements (A - D)

- (A) Withdrawal syndrome is characterised by anxiety, Shakiness, Nausea and Sweating.

- (B) Dependence leads the patient to ignore all social norms in order to get sufficient funds to satiate his/her needs.
- (C) Addiction is a psychological attachment to certain effects associated with drugs and alcohol.
- (D) The immediate adverse effects of drugs and alcohol abuse are manifested in the form of respiratory failure, facial acne, enlarged prostate, and vandalism.

How many of the above statements are true ?

- (1) 3
- (2) 1
- (3) 4
- (4) 2

170. Given below are two statements:

Statement-I : RNA interference takes place in all Eukaryotic organisms as method of cellular defense.

Statement-II : RNAi involves the silencing of a specific mRNA due to a complementary single stranded RNA molecule that binds and prevents translation of mRNA .

In the light of the above statements, choose the correct answer from the options given below.

- (1) Statement-I is true but Statement-II is false
- (2) Statement-I is false but Statement-II is true
- (3) Both Statement-I and Statement-II are true
- (4) Both Statement-I and Statement-II are false

171. Select the correct statements.

- A. In the 24 hour average duration of cell cycle of a human cell, cell division proper lasts for only about an hour.
- B. During the G₂ phase, proteins are synthesised in preparation for mitosis while cell growth stops.
- C. The chromosomal material becomes untangled during the process of chromatin condensation.
- D. Cell-plate represents the middle lamella between the walls of two adjacent human cells.
- E. Cell growth results in disturbing the ratio between the nucleus and the cytoplasm

- (1) A, B, C, D & E
- (2) A, C, D & E only
- (3) A, D & E only
- (4) A, C & E only

172. Read the following statements :-

- (A) Cyclosporin A is used as a clot buster
- (B) Whisky, brandy & rum are produced by distillation of fermented broth
- (C) Baculoviruses are used as biological control agents
- (D) The biogas production technology in India is developed by ICAR only

Which of the above statements are not incorrect?

- (1) A & C
- (2) B & C
- (3) C & D
- (4) A & D

173. In which type of natural selection two peaks are observed?

- (1) Stabilizing selection
- (2) Cyclic selection
- (3) Directional selection
- (4) Disruptive selection

174. Match List-I with List-II

List-I	List-II
(A) Genetically engineered Human Insulin	(I) Gene therapy
(B) GM Cotton	(II) E. coli
(C) ADA Deficiency	(III) Antigen-antibody interaction
(D) ELISA	(IV) Bacillus thuringiensis

Choose the answer from the options given below :

- (1) A-III, B-II, C-IV, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-II, B-IV, C-I, D-III

175. **Assertion** : Symptoms of amoebiasis include cramps, stool with excess mucous & blood clots.

Reason : Housefly act as mechanical carriers and serve to transmit parasite from faeces to food.

- (1) Both Assertion and Reason are True and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason is True but Reason is not the correct explanation of Assertion.
- (3) Assertion is True but Reason is False.
- (4) Both Assertion and Reason are False.

176. Given below are two statements :

Statement-I : The Indian Government has set up GEAC which will make decisions regarding the validity of GM research.

Statement-II : Biopiracy is the term used to refer to the use of bio-resources by indigenous or local people.

In the light of the above statements, choose the correct answer from the options given below :

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

177. A number of marsupials, each different from the other evolved from an ancestral stock but all within the _____ continent.

- (1) African
- (2) Indian
- (3) Australian
- (4) South America

178. **Assertion (A)** : Thorax of cockroach is related with flight & walking of cockroach.

Reason (R) : Mid portion of cockroach body called thorax bears wings & legs.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
- (2) Both Assertion and Reason are correct but Reason is NOT the correct explanation of Assertion
- (3) Assertion is correct and Reason is incorrect
- (4) Assertion is incorrect and Reason is correct

- 179.** Animals that have had their DNA manipulated to possess and express a extra (foreign) gene are called :
- (1) Transgenic animals
 - (2) Somatic hybrids
 - (3) Somaclones
 - (4) Carrier animals
- 180.** Two microbes found to be very useful in genetic engineering are
- (1) crown gall bacterium and *Caenorhabditis elegans*
 - (2) *Escherichia coli* and *Agrobacterium tumefaciens*
 - (3) *Vibrio cholerae* and a tailed bacteriophage
 - (4) *Diplococcus* sp. and *Pseudomonas* sp